

High-Level Design (HLD) Document

Adaptive Offline-First Intelligent Data Pre- Fetching System (AOFIDPS)

For Low Connectivity Regions

Table of Contents

1 . Executive Summary	
2 . System Overview	2 . 1 Project Background 2 . 2 Vision 2 . 3 Key Objectives 2 . 4 Scope and Boundaries
3 . Architecture Principles	3 . 1 Design Philosophy 3 . 2 Architectural Patterns 3 . 3 Critical Design Decisions
4 . Technology Stack	
5 . System Architecture	5 . 1 High-Level Architecture Diagram 5 . 2 Core Platform Components
6 . Module Architecture Overview	6 . 1 Educational Content Module 6 . 2 Healthcare Resources Module 6 . 3 Communication Tools Module (EORS Integrated) 6 . 4 Essential Digital Services Module
7 . Data Architecture	
8 . Security Architecture	
9 . Deployment Architecture	
10 . Risk Management	
11 . Deepest Problem-to-Solution Analysis (All Modules)	
12 . Glossary	

1 . Executive Summary

Problem Statement

Low-connectivity and no-signal regions suffer from digital exclusion. Users cannot reliably access: Educational resources - Healthcare guidance - Emergency communication - Government and financial services

Continuous internet dependency renders modern digital services unusable in rural, disaster-prone, and infrastructure-deficient environments.

Proposed System

The Adaptive Offline-First Intelligent Data Pre-Fetching System (AOFIDPS) is a modular distributed m platform that:

- Predicts user needs
- Pre-fetches prioritized content
- Enables offline functionality
- Synchronizes intelligently
- Provides opportunistic emergency communication

This system is built on edge intelligence + distributed networking + cloud-native backend architecture.

2 . System Overview

2 . 1 Project Background

Existing apps assume stable connectivity. Offline support is typically limited and non-intelligent.

Rural and remote populations require: - Predictive caching - Autonomous local decision engines
Opportunistic relay communication - Low-bandwidth optimized sync

2 . 2 Vision

Create a resilient digital ecosystem that operates effectively in intermittent connectivity environments.

2 . 3 Key Objectives

- Reduce internet dependency by > 70 %
- Ensure emergency message latency < 5 seconds when relay available
- Enable fully offline education and healthcare decision support
- Provide staged digital transactions

2 . 4 Scope and Boundaries

In Scope: - On-device AI prediction - Prefetch engine - Secure sync - Emergency relay - Modular services

Out of Scope: - Satellite hardware integration - Replacing telecom infrastructure

3 . Architecture Principles

3 . 1 Design Philosophy

- Offline-first
- Edge intelligence
- Privacy-by-design
- Event-driven architecture
- Modular extensibility

3 . 2 Architectural Patterns

- Edge Computing
- Microservices Backend
- CQRS
- Event Streaming
- Opportunistic Mesh Networking

3 . 3 Critical Design Decisions

- On-device ML inference
- BLE for discovery
- Delta sync only
- Encrypted local storage
- Priority-based data eviction

4 . Technology Stack

Mobile: - Flutter - Dart - Isar DB - TensorFlow Lite - Native BLE APIs

Backend: - FastAPI - PostgreSQL - Redis - Cloud Run - Object Storage

Security: - TLS 1 . 3 - ECC Key Pairs - AES- 2 5 6 - JWT

5 . System Architecture

5 . 1 High-Level Architecture Diagram

Mobile App  Prediction Engine  Local Storage  Sync Engine  Cloud Backend

Parallel Emergency Relay Layer: User → BLE Broadcast → Relay Device → Backend → Dispatch

5 . 2 **Core Platform Components**

- 1 . Prediction Engine
 - 2 . Prefetch Manager
 - 3 . Cache Controller
 - 4 . Sync Engine
 - 5 . Relay Engine
 - 6 . Security Manager
 - 7 . Cloud API Gateway
-

6 . **Module Architecture Overview**

6 . 1 **Educational Content Module**

Core Problem

Students cannot access learning materials during network outages.

Deep Solution

AI-driven predictive content prefetching + modular content packaging.

Architecture

- Content metadata ranking
- Behavioral modeling
- Delta updates
- LRU eviction

Implementation Approach

- Track user subject interactions
- Predict next topics
- Prefetch compressed video/text bundles
- Store locally with TTL

Offline Capabilities

- Full video playback
 - Quiz engine
 - Progress tracking
-

6 . 2 Healthcare Resources Module

Core Problem

No access to doctors or medical databases in remote areas.

Deep Solution

On-device medical decision engine + offline knowledge base.

Architecture

- Decision tree inference engine
- Symptom input module
- Local medical database
- Sync medical logs

Implementation

- Encode WHO protocol rules
 - Use lightweight ML for symptom classification
 - Store anonymized patient logs
-

6 . 3 Communication Tools Module

(Integrated EORS)

Core Problem

No real-time emergency communication without infrastructure.

Deep Solution

Opportunistic encrypted relay network.

Architecture

- BLE discovery
- Multi-hop routing
- Encrypted payload forwarding
- Cloud dispatch integration

Implementation

- Public key encryption
 - UUID-based duplicate suppression
 - Priority routing
-

6 . 4 Essential Digital Services Module

Core Problem

Users cannot complete financial/government tasks offline.

Deep Solution

Transactional staging + secure delayed execution.

Architecture

- Offline form builder
- Local transaction token
- Sync commit protocol
- Conflict resolution logic

Implementation

- Queue financial requests
 - Use idempotent APIs
 - Implement version control
-

7 . Data Architecture

Mobile: - Encrypted Isar DB - Content cache - Action queue - Relay logs

Backend: - User registry - Emergency events - Content metadata - Analytics engine

8 . Security Architecture

- End-to-end encryption
 - Device identity verification
 - Certificate pinning
 - Encrypted local storage
 - DDoS protection
-

9 . Deployment Architecture

- Cloud Run containers
 - Load balancer
 - Regional replication
 - Managed PostgreSQL
 - Redis cache
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1 0 . Risk Management

Risk	Impact	Mitigation
Low adoption	Weak relay network	Community rollout
Storage overload	App bloat	Smart eviction
Battery drain	Poor UX	Adaptive scanning
Security breach	Data leak	PKI encryption

1 1 . Deepest Problem-to-Solution Analysis

Educational: - Predictive caching reduces repeated downloads - Compression algorithms reduce storage footprint

Healthcare: - Offline inference reduces dependency - Protocol-based decision tree ensures safety

Communication: - Multi-hop relay ensures probabilistic delivery - Priority interrupt ensures low latency

Essential Services: - Transaction staging ensures continuity - Idempotent commit prevents duplication

1 2 . Glossary

AOFIDPS - Adaptive Offline-First Intelligent Data Pre-Fetching System EORS - Emergency Opportunistic Relay System BLE - Bluetooth Low Energy CQRS - Command Query Responsibility Segregation TTL To Live

Conclusion

AOFIDPS is a resilient, modular, distributed digital platform engineered for low-connectivity environments. It integrates predictive caching, edge AI, secure synchronization, and opportunistic relay networking to provide education, healthcare, communication, and essential services without continuous internet dependency.